

EEE406 (Power System Analysis)

Tutorial 1

Transmission line parameters, Current and Voltage Relations in Transmission Lines

Lecturer: Dr. Abdul Rashid Abdul Rahman

Email: abdulrashid.abdulrahman@xmu.edu.my

Room: A1-440

DRAR

Inductance of Transmission Line

Question 1

Find the loop inductance and reactance per km of a single phase overhead line consisting of two conductors, each 1.213 cm diameter. The spacing between conductors is 1.25 meters and frequency is 50 Hz.

Inductance of Transmission Line

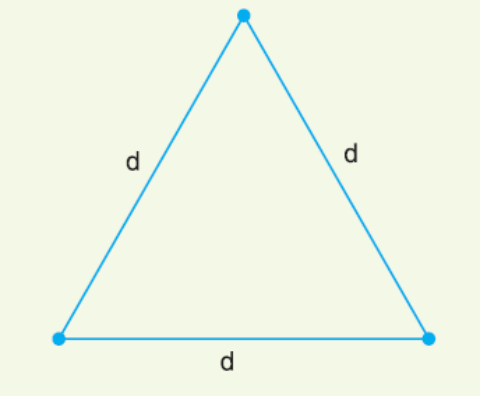
Question 2

A single phase line has two parallel conductors 2 meters apart. The diameter of each conductor is1.2cm. Calculate the loop inductance per km of the line.

Inductance of Transmission Line

Question 3

Find the inductance per km of a 3-phase transmission line using 1.24cm diameter conductors when these are placed at the corners of an equilateral triangle of each side 2m.



DRAR

Question 4

A 138 kV, 200 MVA, 60 Hz, three-phase, power transmission line is 100 km long, and has the following characteristics:

$r = 0.103\ \Omega/\text{km}$

$x = 0.525\ \Omega/\text{km}$

$y = 3.3 \times 10^{-6}\ \text{S/km}$

(a) What is per phase series impedance and shunt admittance of this transmission line?
(b) Should it be modeled as a short, medium, or long transmission line?
(c) Calculate the *ABCD* constants of this transmission line.
(d) Sketch the phasor diagram of this transmission line when the line is supplying rated voltage and apparent power at a 0.90 power factor lagging.
(e) Calculate the sending end voltage if the line is supplying rated voltage and apparent power at 0.90 PF lagging.
(f) What is the voltage regulation of the transmission line for the conditions in *(e)*?
(g) What is the efficiency of the transmission line for the conditions in *(e)*?